25900333 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCCFS201 - COMPUTER ORGANIZATION

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **1** marks

1. Name the special purpose register that holds the address of the memory location to be accessed [K] / [CO1]
2. Define the term "conditional branch" [K] / [CO1]
3. Recognize the part of a computer system that tells the computer what to do [K] / [CO1]
4. Define the term "bus" [K] / [CO1]
5. Illustrate the purpose of the PC (Program Counter). [An] / [CO2]
6. Inspect the role of SP in locating the top of the stack. [An] / [CO2]
7. Identify the memory which is called as auxillary memory. [U] / [CO1]
8. Examine why hard disks are preferred for storing operating systems and application softwares. [An] / [CO2]
9. Indicate the two types of cache memory. [U] / [CO1]

10. State why ROM is required for booting. [K] / [CO2]

11. Examine why cache memory is placed between CPU and main memory. [An] / [CO2]

12. State why does DRAM better suited for larger memory modules.. [K] / [CO2]

[1x10 = 10]

Part B

Short Essay Type Questions

Answer any **4** questions

Each question carries **5** marks

13. Discuss the differences between load and store memory operations. [U] / [CO1]

14. Describe the functions of various registers in a CPU. [U] / [CO1]

15. Explain the function of auto-increment addressing mode with suitable example. [U] / [CO1]

16. Compare and contrast Relative and Indexed addressing modes with suitable examples. [An] / [CO2]

17. Describe the functions of Processing Unit in a computer system. [U] / [CO1]

18. Discuss the importance of instruction register in fetch cycle. [U] / [CO1]

[5x4 = 20]

Part C

Essay Type Questions

Answer any **2** questions

Each question carries **10** marks

19. Describe the importance of different functional units of a computer system with the help of a block diagram [U] / [CO1]

20. Discuss in detail about data transfer and data manipulation instructions. [U] / [CO1]

21. Differentiate between arithematic operations and logic operations with example. [An] / [CO2]

[10x2 = 20]